

N 33

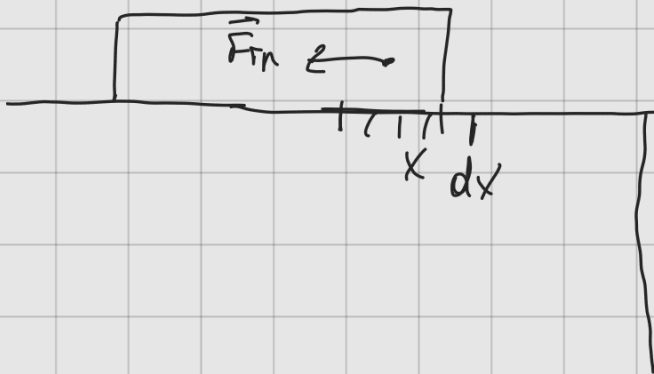
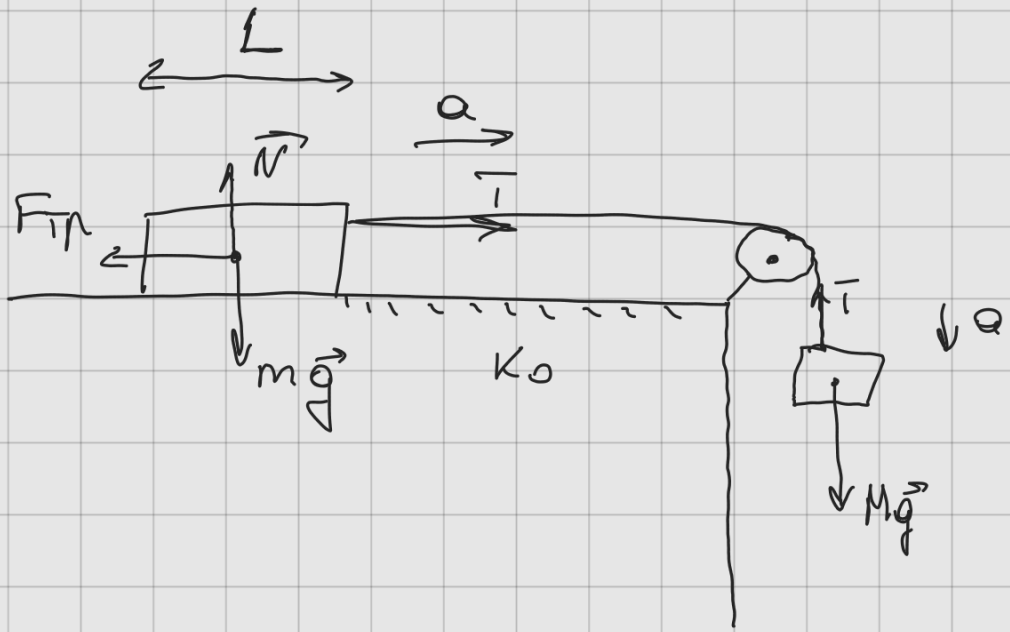
Дано:

Решение:

M, L, m

$\kappa = \kappa_0$

Найти: v



$$F_{тр} = \frac{x}{L} \mu mg$$

$\frac{x}{L} m$ - масса массов

$$\Delta W = A_{н.с.}$$

$$-MgL + \frac{(M+m)v^2}{2} = A_{н.с.}$$

$$\int \delta A = \int_0^L -\frac{x}{2} \mu mg dx$$

$$A = -\frac{x^2}{2 \cdot 2} \mu mg \Big|_0^L \rightarrow A = -\frac{L^2}{2} \mu mg = -\frac{\mu mg L^2}{2}$$

$$v = \sqrt{\frac{2}{M+m} (MgL - \frac{\mu mgL}{2})} = \sqrt{\frac{2gL}{M+m} (M - \frac{\mu m}{2})}$$

N34



Теорема Кёнига:

$$K_{\text{пол}} = \frac{m v_0^2}{2} + \frac{\frac{m}{3} v_0^2}{2} = \frac{2}{3} m v_0^2 = \frac{(m v_0 - \frac{m v_0}{3})^2}{2 \cdot \frac{4}{3} m} + K_{\text{цм}}$$

$$p_{\text{цм}} = m v_0 - \frac{m v_0}{3} = \frac{2}{3} m v_0$$

$$K_{\text{цм}} = \frac{\frac{4}{9} m^2 v_0^2}{2}$$

$$\frac{2}{3} m v_0^2 = K_{\text{цм}} + \frac{\frac{4}{9} m^2 v_0^2}{\frac{8}{3} m} = K_{\text{цм}} + \frac{\frac{4}{9} m v_0^2 \cdot \frac{3}{8}}{\frac{2}{2}} = K_{\text{цм}} + \frac{m v_0^2}{6}$$

$$K_{\text{цм}} = \frac{m v_0^2}{2}$$

Кинетическая энергия ц.м. не может быть изменена внутренними силами

X

3

1

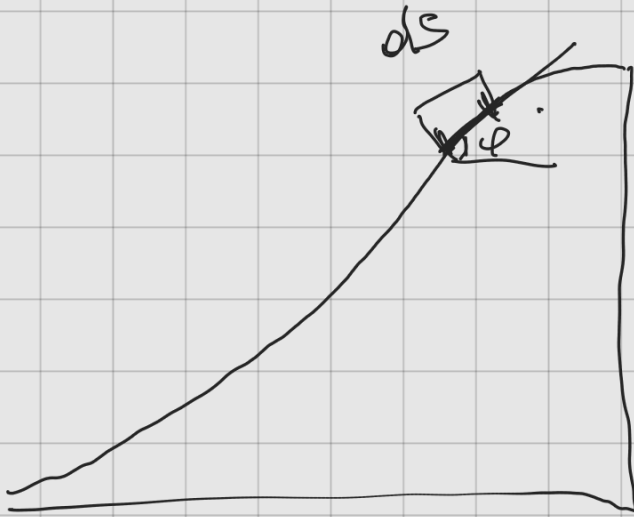
5

$$\Delta \Pi = -2 \int_0^x k x^{\frac{5}{2}} dx = \frac{4}{5} k x^{\frac{5}{2}}$$

$$\frac{m v_0^2}{2} = \frac{4}{5} k x^{\frac{5}{2}} \Rightarrow x = \left(\frac{5 m v_0^2}{8 k} \right)^{\frac{2}{5}}$$

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a)



$$N = mg \cos \varphi$$

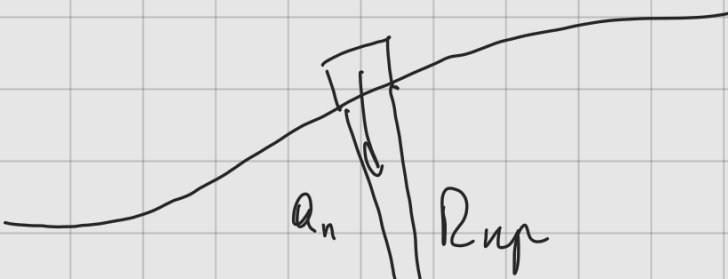
$$F_{fr} = \mu mg \cos \varphi$$

$$\int_0^l \delta A_{fr} = - F_{fr} ds = - \underbrace{\mu mg \cos \varphi}_{de} ds = \int_0^e - \mu mg de$$

$$A_{fr} = - \mu m g l$$

$$\Delta \Pi = A_F + A_{fr} \Rightarrow m g h + \mu m g l = A_F = m g (h + \mu l)$$

b)



$$a_n = \frac{v^2}{R_{kr}}$$

$$N = mg \cos \varphi - m a_n = mg \cos \varphi - \frac{m v^2}{R_{\text{кр}}}$$

$$\delta A_p = -\int N ds = -\int mg \cos \varphi ds + \int m \frac{v^2}{R_{\text{кр}}} ds =$$

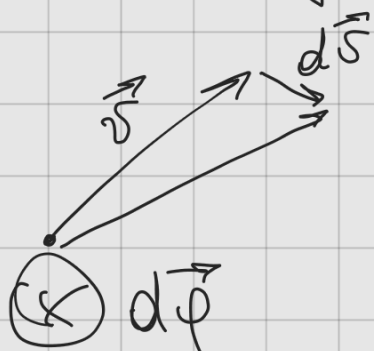
$$= -\int mg dl + \int m v^2 d\varphi$$

$$\frac{ds}{R_{\text{кр}}} = d\varphi$$

$$\Delta \Pi + \Delta K = mgh = A_F + \int (-\int mg dl + \int m v^2 d\varphi)$$

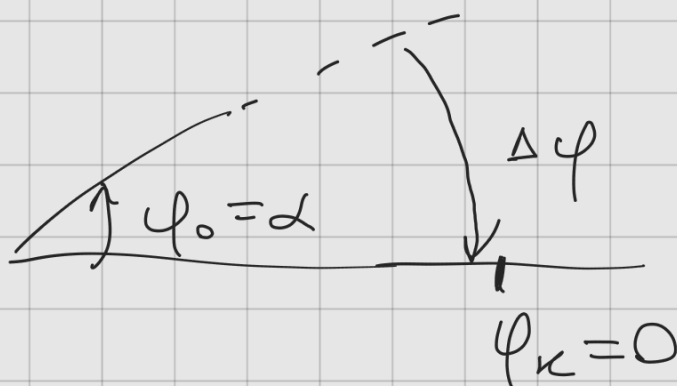
$$mgh = A - \int mg dl + \int m v^2 d\varphi \Big|_{-\alpha}^0 = A - \int mg dl + \int m v^2 d\varphi$$

$$\Rightarrow A = m (gh + \int g l - \int v^2 d\varphi)$$



$$d\vec{s} = d\vec{\varphi} \times \vec{v}$$

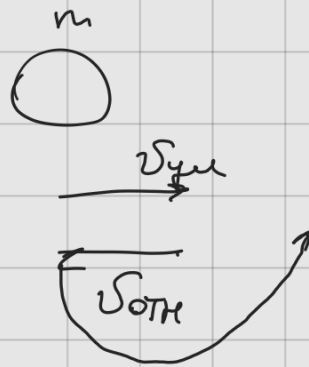
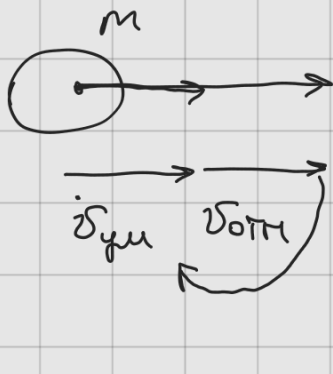
(правильно
направление
руки)



$\Delta \vec{\varphi}$ от нач

$d\vec{\varphi} \uparrow \vec{v} \Rightarrow$ значение угла положительно

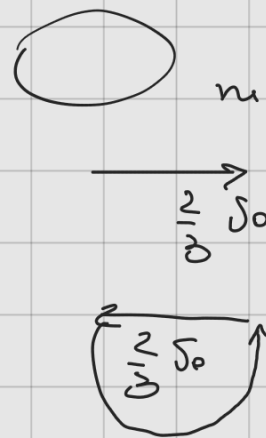
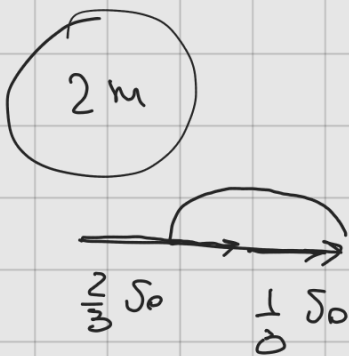
N36



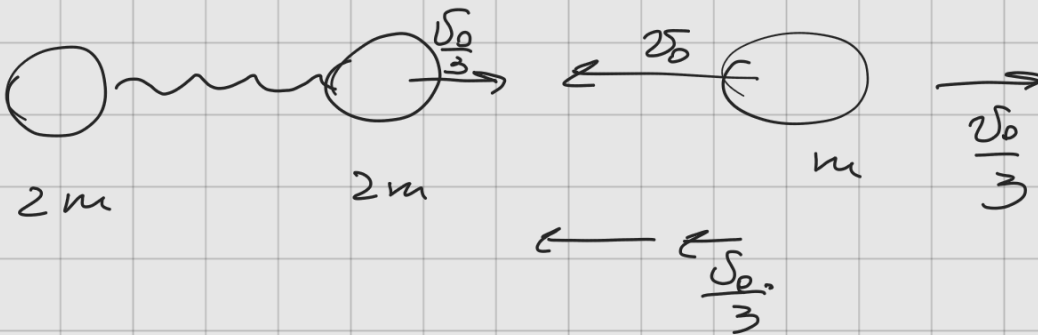
$$v_{cm} = \frac{m_1 v_{1x} + m_2 v_{2x}}{2}$$

$$= \frac{m \cdot v_0 + m \cdot 0}{2m} = \frac{v_0}{2}$$

Сохраняется относительные скорости



$$\frac{2v_0}{3}$$



$$0 + \frac{2m \left(\frac{2v_0}{3} \right)^2}{2} = \frac{\left(\frac{4v_0 m}{3} \right)^2}{2 \cdot 4m} + \frac{kx^2}{2}$$

$$\frac{8}{9} m v_0^2 = \frac{4}{9} m v_0^2 + \frac{1}{2} k x^2$$

$$x = \frac{2}{3} J_0 \sqrt{\frac{m}{k}}$$
